



Generative AI in Retail & CPG

Strategic Impact on Personalization, Supply Chain & Customer Experience

What is Artificial Intelligence?

Definition

Machines performing tasks requiring human intelligence — learning, reasoning, decision-making

3 Types

Narrow AI (Siri, Alexa) • **General AI** (in development) • **Super AI** (theoretical)

Coined 1956

John McCarthy defined AI as "the science and engineering of making intelligent machines"



Retail & CPG: Industry Evolution

From ancient barter systems to AI-powered smart stores — the retail industry is one of the most dynamic sectors in the world.



CPG Leaders Using AI

- Unilever, Procter & Gamble, Nestlé, PepsiCo
- Focus: demand forecasting, personalized marketing, smart supply chains

AI Adoption in Retail

Rapid growth from 2020–2025 across inventory management, recommendation systems & customer engagement

Generative AI: Applications in Retail & CPG



Personalized Recommendations

AI analyzes browsing history & purchase behavior to suggest relevant products



Chatbots & Virtual Assistants

24/7 customer support, instant query resolution, order tracking



Inventory & Demand Forecasting

Predictive analytics reduces overstocking, shortages & operational costs



Supply Chain Optimization

Route planning, warehouse automation, real-time monitoring



Dynamic Pricing & Fraud Detection

AI adjusts prices in real time; identifies suspicious transactions



Personalized Marketing

AI-generated ads, promotions & product descriptions at scale

AI Adoption: Key Industry Players



Amazon

AI-powered recommendations, Alexa, warehouse automation & demand forecasting



Sephora

Virtual try-on, AI beauty recommendations & personalized in-store experiences



Walmart

Predictive analytics for inventory, AI chatbots & real-time supply chain decisions



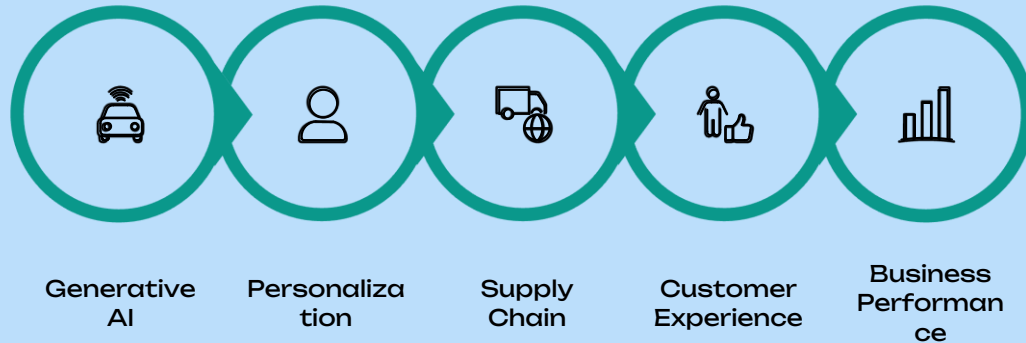
Flipkart

AI for recommendations, fraud detection, delivery optimization & customer support



Research Framework & Objectives

Conceptual Framework



Study Objectives

- Analyze Generative AI's impact on customer personalization
- Evaluate AI in inventory, forecasting & supply chain
- Identify benefits, challenges & ethical considerations
- Assess AI's role in digital transformation & competitive advantage

Research Methodology

Descriptive

AI adoption trends,
customer behavior &
digital commerce patterns

Analytical

Impact of AI on business
performance &
operational efficiency

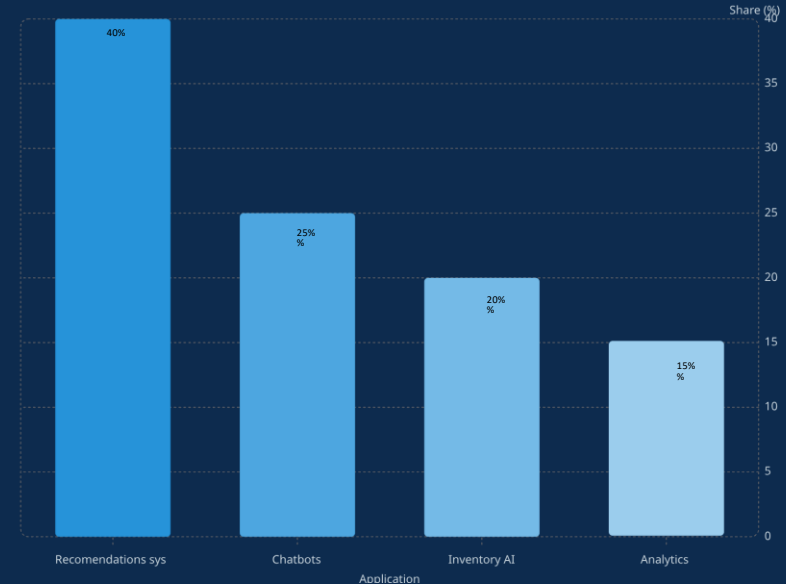
Quantitative

Google Forms surveys,
statistical analysis via
Microsoft Excel

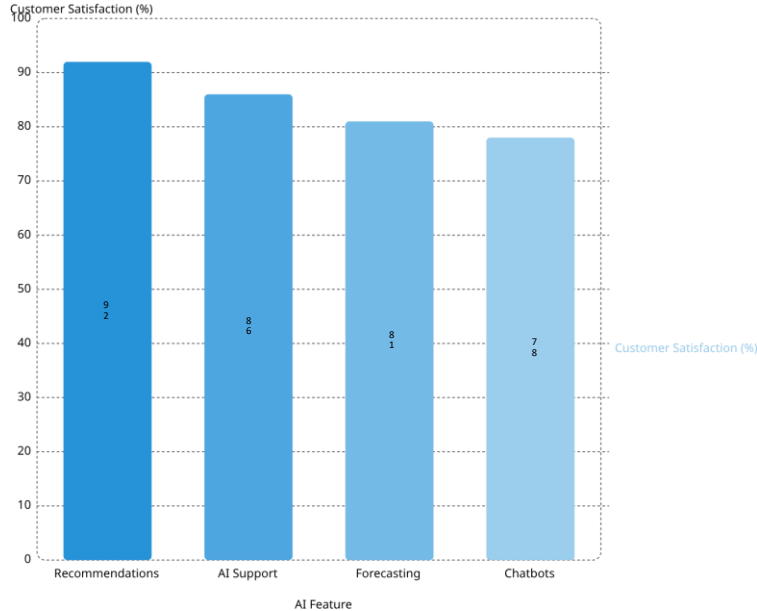
Qualitative

Customer interviews, case
studies & secondary
industry reports

AI Applications in Retail — Distribution



Data Analysis: Key Findings



Customer satisfaction with AI features — Recommendations lead at 92%

Major Findings

Personalization

Customers strongly prefer AI-driven product recommendations

Chatbots

24/7 support improves trust, retention & response speed

Supply Chain AI

Predictive analytics reduces delays & inventory shortages

Competitiveness

AI adopters outperform traditional retail systems

Challenges & Recommendations

Challenges Identified

- Data privacy & cybersecurity risks
- High implementation costs
- Employee resistance to automation
- Ethical AI & algorithmic bias concerns
- Technology dependence & system failure risks

Strategic Recommendations

- Invest in recommendation systems, predictive analytics & segmentation tools
- Strengthen cybersecurity & transparent data privacy policies
- Launch employee training & AI upskilling programs
- Follow ethical AI practices — fairness, transparency, accountability

Conclusion & Future Scope

Generative AI is the most important driver of personalization, efficiency & competitiveness in Retail & CPG. Early adoption of AI will help organizations build stronger customer loyalty and achieve a sustainable market advantage.

- Organizations that invest in AI and responsible digital transformation are best positioned for sustainable success in the evolving digital economy.



Thank You

We appreciate your time and attention.

